

Exam Instructions

Statistical Inference and Learning

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Computer Science and Information Technology

Project Proposal

Achievement of the course objectives through the oral discussion of a project **agreed with the teacher**

The project consists in the **individual and original analysis** of a dataset

Focus must be given to the **methods learned** in the course

Comparisons with methods learned in other courses are welcome, but such comparisons cannot replace the methods learned in the course

The student is required to:

- prepare a **short report**
- give a **presentation** and **discuss** the report

Project Proposal

Each student must submit a [project proposal](#) in advance (Moodle form)

The project proposal should include:

- the description of the data
- the data source, typically an URL address
- the purpose of the project
- how the project differs from the analyses of the same data already available (if any)

Students can work on their project only [after the teacher has accepted the project proposals](#)

Materials required

The project must be submitted **at least one week before** the oral examination

Materials to be submitted:

1. **short report** of maximum 10 pages (12 point font, doublespace)
2. documented code written in **R Markdown** (.Rmd and compiled .html)

The report should:

- describe the steps of the statistical analyses
- provide a detailed interpretation of results
- include illustrations of the results

The documented code must allow to reproduce the results completely

Code written in languages other than R will **not be accepted**

Presentation and Discussion of the Project

On the day of the exam, an oral discussion of the project will be held

Each student has 15 minutes to give a presentation (slides) of their project

Discussion of the project will concern:

- technical aspects of the project
- substantial (interpretative) aspects of the project

Please note: during the discussion students may also receive questions about course topics not covered in the project!

Evaluation

Students will be evaluated in terms of

- originality of the project
- quality of the statistical analyses
- correct use of the technical terminology
- correct conclusions
- quality of the presentation
- quality of the oral discussion

The teacher will check for previous analyses of the same data and how the project differs from them

It might be okay if there is a previous “similar” analysis if the project work is sufficiently different or deeper in some respects to be considered **original**

Failures or Declinations

If the student fails the exam, they can try another session with the **same** project

However, if the exam is failed again, then a **new** project must be considered for the subsequent exam sessions

If the student passes the exam but decides to decline the score, then a **new** project must be considered for the subsequent exam sessions

The new project must be agreed with the teacher as for the initial project and follow the same steps

Exam sessions are not possible outside the four official dates

Office Hours

You can discuss with me technical aspects of your project such as problems with R commands during office hours

Please do **not** email me help requests about your code

I will not reply to those emails as they will likely generate long and confusing discussions

Instead, come to my office during the dedicated office hours or ask for a Zoom meeting if you cannot come to my office in person

Questions?

